

1.1: Solving Problems:

There are 4 Stages and a pretty diagram to solving every problem EVER:

1. Read the question & identify what you are being asked to find out.
2. Picking out all relevant information from the question
3. Applying the appropriate mathematical technique to the info in the question
4. Check your answer makes sense within the question's context.

1.2: Mathematical Notation:

$\Rightarrow$: Leads to / Implies

$\Leftarrow$: Follows from / is implied by.

$\Leftrightarrow$: If something both implies & is implied by (iff)

Necessary & Sufficient:

$A \Rightarrow B$ · A is a sufficient condition for B.

$A \Leftarrow B$ · A is a necessary condition for B.

$A \Leftrightarrow B$ · A is both necessary & sufficient.

The Converse:

$A \rightarrow B$

Converse: $B \rightarrow A$.

1.3: Proof:

You can prove a mathematical statement (a conjecture) to be true / false, by using rigorous methods.

Proof By Exhaustion:

Logical argument which explains why a conjecture is true & often requires the use of algebra to ensure that it is true for all values.

Proof by Exhaustion:

Sometimes it will be necessary to test all possible cases to prove that a statement is true for all values.

Some Conjectures can be disproved using a single counter example.